
Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that conver-
4 gence is concentrated where the embedding manifold is flat. A pointwise curvature
5 field from a trained decoder, validated against analytic truth, finds mean-curvature
6 magnitude correlating negatively with cross-survey neighbour agreement at every
7 latent dimension, and that local density removes most of it ($d = 20$), half ($d = 25$)
8 or little of an already small value ($d = 32$). A second-order expansion of a linear
9 map restricted to the manifold explains why: a readout sees the second fundamen-
10 tal form contracted with its normal component, $\langle w_N, \Pi \rangle$, and, at second order, an
11 alignment’s error splits into a connection mismatch and the relative second funda-
12 mental form projected into the target normal space; mean curvature retains only
13 the metric trace of Π and determines neither. On a unit-normalized embedding
14 that Hessian splits into a sphere term proportional to the readout’s value and a
15 shape term. On a synthetic surface the sphere term holds at -0.74 to -0.80 across
16 samplings that swing the $\|H^S\|$ partial from -0.29 to $+0.26$; on 86,471 labelled
17 galaxies the mismatch between the label Hessian and the full probe-facing Hessian
18 is negative wherever the label has local structure and survives cross-fitting and a
19 change of encoder, intervening on the readout shows the manifold’s own bending
20 raises local R^2 at nearly every anchor while its mirror image lowers it, the sphere
21 term is negative on two of four labels under a weak-ridge sensitivity, and the shape
22 term alone takes a label-dependent sign consistent with that decomposition. Mean
23 curvature of a single representation is therefore not a viable explanation for weak
24 Platonic convergence; the obstruction is relative and, for the fitted global linear
25 alignment, already present at first order.

26 1 Introduction

27 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
28 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
29 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by
30 a family of different vision models share only 0.3 to 2.3 percent of their nearest neighbours across
31 the two surveys. We ask where on the manifold that agreement lives: if two models disagree most
32 where the embedding surface bends sharply, local mean curvature should associate negatively with
33 local cross-survey agreement.

34 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
35 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled

test of the curvature–alignment prediction, which finds no effect free of density; (iii) the residual expansion of a linear map on a submanifold, which names the curvature such a map responds to and shows mean-curvature magnitude is blind to it; and (iv) known-surface and real-data tests on the readout side, where the expansion’s terms behave as predicted.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], (Platonic Universe release [Duprat et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. Embeddings are unit-normalized (data on S^{767}). Curvature is evaluated only where J has full rank.

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$, by automatic differentiation. On the unit sphere every d -submanifold carries a radial component of H equal to $-d$ that says nothing about shape. To isolate shape, the map differentiated is $F/\|F\|$, whose H splits exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}); H^S , normal to M within the sphere, carries the shape. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the projected decoder.

Validation. Scored against closed-form curvature on rank (Spearman), direction (cosine) and magnitude ratio. At $D = 768$ an in-sphere surface $G(z) = Q \text{ normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four Gaussian bumps, Q a rotation; 18-dimensional span) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H^S\|$ (equivalently $\|H\|$, the radial part being fixed), direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

3 Curvature and cross-survey convergence

Probe. Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local, correlated with $\|H_{\text{raw}}\|$ over all 10,000 points at $d \in \{20, 25, 32\}$.

Result. The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009), in the predicted direction. But k -NN statistics are density-sensitive [Radovanović et al., 2010]: local density (the $k = 30$ nearest-neighbour estimate on the Legacy embedding, $d = 20$) correlates with agreement at -0.21 at every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 : 78%, 49% and 8% of the association removed. Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; three decoder seeds; the CKA [Kornblith et al., 2019] tertile gap agrees). Removing the radial component of H_{raw} (the raw surface is off the sphere; the two rank at $\rho = 0.89$ – 0.96) collapses the $d = 25$ partial to -0.023 , into the range of the two d that fail. Convergence is not detectably concentrated by mean-curvature magnitude once density is controlled.

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$, so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

To second order, a linear function is as non-linear on M as M bends in that function’s normal direction [Lee, 2018]. For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the rest, $\Pi = \Pi^S - g\hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x})g$ (2): a linear readout of a unit-normalized

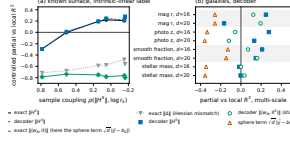


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the sample's density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$: partial under the multi-scale density control for the decoder's $\|H^S\|$ (filled squares), the shape term (open circles) and the sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

embedding is bent by the manifold's shape and, whatever the manifold, by the sphere in proportion to the readout's own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 (the $\frac{1}{2}$ is the Gaussian constant; real k -NN patches share only the tensor structure), with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A \Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear alignment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter when the induced connections agree at x_0 (in particular under a local intrinsic isometry); the extrinsic term's normal part is $\Pi_G - P_N^G A \Pi_F$.

So $\|H\|$, the norm of the metric trace of Π , need not correlate with either $\langle w_N, \Pi \rangle$ or the relative term; the second-order term scales as s^4 and, with fixed- k neighbourhoods, density sets s , so curvature effects scored on patches are density-sensitive.

5 The probe-facing curvature, on a known surface and on galaxies

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors' 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman-Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at $k \in \{16, \dots, 2048\}$ (five scales). Per anchor we compute $\|H^S\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$ with $w_N = w - Jg^{-1}J^\top w$, split by Eq. (2) into its shape and sphere terms.

Known surface. The validation surface is resampled with weight $\|H^S\|^\gamma$, so only where the points sit changes (Figure 1a). The $\|H^S\|$ partial follows the sampling from -0.29 to $+0.22$, and the decoder reproduces it. The probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ has partial correlation -0.74 to -0.80 at every sampling. On this surface it is the sphere term of Eq. (2) (rank 0.996–1.000 with $\sqrt{d}|\hat{y} - b_0|$).

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours (a local Monge chart, where the Christoffel symbols vanish at the anchor, so the quadratic

coefficient is the covariant Hessian). Three regularities (Table 1; other decoder seeds and widths in Appendix A). The sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass, where the sealed-probe sign came with ridge shrinkage of extreme predictions. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is indistinguishable from zero throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). Writing $K = \langle w_N, \Pi \rangle$, curvature lowers the second-order residual relative to $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle > \|K\|^2$; positive alignment is the first-order condition; the mismatch column is the full test. Intervening on the readout confirms the sign at the anchor level: with the probe’s in-sphere normal component replaced by the decoder’s second-order term $\frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$, local R^2 rises at 79–99% of anchors, falls at 97–100% when that term is mirrored, and a random normal direction of the same norm does neither (Appendix E; five encoders). The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift), consistent with Eq. (3): it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, K \rangle + \|K\|^2$, so raising it shrinks or enlarges the mismatch according to that condition. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55 – 0.65), and an independent split-half quadratic-chart estimator of Π gives the same shape-term sign on three of four labels.

Across encoders. The quadratic-chart estimator’s mean-bending trace on five encoders at the same anchors (ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$, $k = 2,048$) has density-controlled associations with the magnitude probe’s local R^2 between -0.24 and $+0.27$, of both signs: a trace statistic has no encoder-stable sign, consistent with Eq. (1). The decoder test repeated per encoder (Appendix D) is encoder-stable where the expansion says it should be: the mismatch partial is negative for magnitude and redshift on all five encoders (-0.32 to -0.61 ; cross-fitted -0.28 to -0.62) and the alignment positive for magnitude on all five ($+0.13$ to $+0.53$) and for redshift on four; the shape term’s sign varies with encoder as well as label; stellar mass stays null but for one mismatch cell (-0.12).

6 Discussion

A decoder yields a pointwise second fundamental form at production scale that agrees with an independent estimator and with the data. The residual expansion names what a linear map responds to, and on galaxies its terms behave as predicted: the Hessian mismatch carries the predicted negative sign, robust to cross-fitting, probe strength and the choice of encoder; the sphere term does so on two of four labels under a weak-ridge sensitivity; and the shape term is label-signed because it enters through its alignment with, and magnitude relative to, the label’s own Hessian. The same expansion explains why mean-curvature statistics disagree across estimators and encoders: they are traces of a tensor that each map contracts differently. If weak convergence is attributed to the failure of a linear alignment map, one embedding’s mean curvature is not the relevant second-order quantity; after tangent alignment the relevant normal term is $\Pi_G - P_N^G A \Pi_F$, which needs both embeddings’ second fundamental forms and a map between tangent frames. Our pilot (Appendix C) finds a 44% first-order residual in carrying HSC tangent frames into the Legacy embedding under the fitted global ridge map, and no estimator-stable second-order association. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10 – 0.17); its norm is reliably ranked (split-half ρ 0.92) but its direction only moderately (tensor cosine 0.19 – 0.35). On a synthetic surface with known curvature the mismatch partial reproduces (-0.14 to -0.88) but the alignment partial has no stable sign; the intervention of Appendix E, not the fixture, is what ties alignment to local R^2 . Stars are per cell; mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

References

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://>

164 //arxiv.org/abs/2405.07987.

165 Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The platonic universe: Do foundation
166 models see the same sky? In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*, 2025.
167 URL <https://arxiv.org/abs/2509.19453>.

168 Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, et al. DINOv3. *arXiv*
169 *preprint*, 2025. URL <https://arxiv.org/abs/2508.10104>.

170 Hiroaki Aihara, Nobuo Arimoto, Robert Armstrong, et al. The Hyper Suprime-Cam SSP Survey: Overview
171 and survey design. *Publications of the Astronomical Society of Japan*, 70(SP1):S4, 2018. doi: 10.1093/pasj/
172 psx066.

173 Arjun Dey, David J. Schlegel, Dustin Lang, et al. Overview of the DESI Legacy Imaging Surveys. *The Astro-*
174 *nomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

175 Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large obser-
176 vation models for astronomy. *arXiv preprint*, 2024. URL <https://arxiv.org/abs/2405.14930>.

177 Miloš Radovanović, Alexandros Nanopoulos, and Mirjana Ivanović. Hubs in space: Popular nearest neighbors
178 in high-dimensional data. *Journal of Machine Learning Research*, 11:2487–2531, 2010.

179 Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network rep-
180 resentations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*,
181 2019.

182 John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer,
183 2nd edition, 2018. doi: 10.1007/978-3-319-91755-9.

184 Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016.
185 arXiv:1610.01644.

186 David Freedman and David Lane. A nonstochastic interpretation of reported significance levels. *Journal of*
187 *Business & Economic Statistics*, 1(4):292–298, 1983. doi: 10.1080/07350015.1983.10509354.

188 Appendix A: decoder ablations

189 Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two
190 further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach
variance explained 0.952). Every significant sign of the main table is retained.

label	quantity	seed 0	seed 1	seed 2	400^3
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	−0.26	−0.26	−0.27	−0.26
	mismatch	−0.39	−0.35	−0.34	−0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	−0.06*	−0.09*	−0.09	−0.06*
	sphere	−0.33	−0.35	−0.34	−0.35
	mismatch	−0.45	−0.43	−0.42	−0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	−0.13	−0.15	−0.14	−0.11
	mismatch	−0.09*	−0.05*	−0.08*	−0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	−0.06*	+0.05*	+0.09
	sphere	−0.35	−0.35	−0.35	−0.35
	mismatch	−0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	−0.02*	+0.03*	−0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

Appendix B: sensitivity of the mismatch and sphere columns

Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	-0.38	-0.34	+0.33	+0.31	+0.35	-0.21 (-0.26)
mag_r	20	-0.43	-0.40	+0.39	+0.37	+0.24	-0.22 (-0.26)
photo_z	16	-0.42	-0.36	+0.23	+0.20	+0.32	+0.05* (-0.33)
photo_z	20	-0.42	-0.42	+0.22	+0.22	+0.20	+0.03* (-0.36)
smooth_fraction	16	-0.10	-0.17	+0.17	+0.19	+0.33	-0.12 (-0.13)
smooth_fraction	20	-0.15	-0.21	+0.05*	+0.03*	+0.23	-0.13 (-0.15)
stellar_mass	16	+0.01*	-0.04*	-0.03*	+0.06*	+0.31	-0.02* (-0.35)
stellar_mass	20	-0.06*	-0.07*	+0.01*	-0.00*	+0.19	-0.02* (-0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

Appendix C: relative second fundamental form, pilot

Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ; median first-order obstruction $\|AJ_F - J_G L\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of each decoder's $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$, the same with L fitted on 256 neighbours' latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F 's tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	-0.15	+0.04*	-0.18	+0.04*	-0.18	+0.13
$d = 25$, seed 0	0.64	0.43	-0.10	+0.06	-0.16	+0.05	-0.23	+0.13
$d = 20$, seed 1	0.64	0.44	-0.11	+0.05	-0.19	+0.04	-0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$, multi-scale density control. The first row per label is the probe's global out-of-sample R^2 ; Table 6 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per encoder.

Appendix E: intervening on the readout's normal component

The partials of Table 1 compare anchors with one another and cannot say whether bending away from the label's curvature hurts relative to no bending, since no anchor is flat. Here the readout is changed at a fixed manifold. At each anchor the fitted ridge weight is split into tangent, radial and

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	-0.11	+0.16	+0.16	-0.21
	sphere	-0.26	-0.23	-0.40	-0.24	-0.08*
	mismatch	-0.39	-0.57	-0.34	-0.32	-0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	-0.06*	-0.05*	-0.17	+0.29	-0.16
	sphere	-0.33	-0.21	-0.29	-0.46	-0.44
	mismatch	-0.45	-0.61	-0.45	-0.48	-0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	-0.08*	+0.20	+0.24	+0.06*
	sphere	-0.13	-0.06*	+0.05*	-0.15	-0.11
	mismatch	-0.09*	-0.33	-0.25	-0.20	-0.01*
	alignment	+0.19	-0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	-0.15	-0.10	-0.11	-0.13
	sphere	-0.35	-0.26	-0.38	-0.40	-0.38
	mismatch	-0.04*	+0.03*	-0.12	-0.06*	-0.02*
	alignment	+0.01*	-0.05*	-0.01*	+0.07*	-0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	-0.60	-0.62	+0.50	+0.50	+0.35
	photo_z	-0.56	-0.59	+0.28	+0.27	+0.24
	smooth_fraction	-0.32	-0.29	-0.03*	-0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	-0.03*	-0.07*	+0.22
CLIP-B	mag_r	-0.28	-0.35	+0.28	+0.29	+0.31
	photo_z	-0.41	-0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	-0.22	-0.24	+0.15	+0.20	+0.28
	stellar_mass	-0.14	-0.13	-0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	-0.36	-0.34	+0.19	+0.21	+0.29
	photo_z	-0.46	-0.45	+0.18	+0.19	+0.30
	smooth_fraction	-0.15	-0.19	-0.02*	+0.03*	+0.32
	stellar_mass	-0.05*	-0.01*	+0.14	-0.03*	+0.29
ViT-L	mag_r	-0.46	-0.49	+0.09*	+0.15	+0.30
	photo_z	-0.44	-0.42	+0.09*	+0.13	+0.24
	smooth_fraction	-0.03*	-0.04*	+0.11	+0.07*	+0.30
	stellar_mass	-0.06*	+0.01*	-0.04*	-0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

219 in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$, and the readout $w_T + w_{\text{rad}} + t q$ is scored on the
 220 anchor's 2,048 neighbours with the local intercept refit, where q is the decoder's second-order term
 221 $\frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$ in the anchor's chart coordinates: $t = 1$ is the manifold bending as it does in the
 222 probe's normal direction, $t = -1$ the mirror-image bending, $t = 0$ flat. The local sum of squares is
 223 an exact quadratic in t with minimum t^* . Columns: fraction of anchors where $t = 1$ beats flat (help)
 224 and where $t = -1$ is worse than flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ;
 225 then the same for a random in-sphere normal direction of the same norm as w_S .

run	label	decoder second-order term					random normal direction				
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	-0.051	2.3	0.28	0.71	-0.006	-0.006	
	photo_z	0.99	0.99	+0.028	-0.038	3.0	0.38	0.70	-0.003	-0.004	
	smooth_fraction	0.90	1.00	+0.021	-0.040	1.7	0.24	0.79	-0.008	-0.009	
	stellar_mass	0.96	0.99	+0.026	-0.040	2.6	0.34	0.68	-0.004	-0.005	
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	-0.049	2.6	0.30	0.72	-0.005	-0.006	
	photo_z	0.99	1.00	+0.029	-0.038	3.6	0.37	0.68	-0.002	-0.004	
	smooth_fraction	0.90	1.00	+0.020	-0.035	1.9	0.26	0.77	-0.007	-0.008	
	stellar_mass	0.97	1.00	+0.027	-0.038	3.1	0.31	0.72	-0.003	-0.004	
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	-0.031	2.0	0.36	0.70	-0.003	-0.004	
	photo_z	0.96	0.99	+0.020	-0.030	2.6	0.36	0.62	-0.002	-0.002	
	smooth_fraction	0.93	1.00	+0.024	-0.055	1.5	0.28	0.72	-0.009	-0.008	
	stellar_mass	0.95	1.00	+0.016	-0.028	2.1	0.32	0.64	-0.002	-0.002	
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	-0.031	2.2	0.34	0.66	-0.004	-0.004	
	photo_z	0.99	1.00	+0.019	-0.026	3.4	0.39	0.62	-0.002	-0.002	
	smooth_fraction	0.92	0.99	+0.014	-0.027	1.7	0.28	0.69	-0.004	-0.004	
	stellar_mass	0.97	1.00	+0.020	-0.028	2.8	0.36	0.65	-0.002	-0.002	
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	-0.031	2.0	0.32	0.64	-0.004	-0.003	
	photo_z	0.99	1.00	+0.018	-0.026	2.9	0.36	0.59	-0.002	-0.001	
	smooth_fraction	0.88	0.98	+0.016	-0.036	1.6	0.31	0.69	-0.004	-0.004	
	stellar_mass	0.99	1.00	+0.022	-0.032	3.1	0.38	0.57	-0.002	-0.001	
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	-0.034	2.6	0.32	0.69	-0.003	-0.004	
	photo_z	0.97	0.99	+0.018	-0.025	3.2	0.40	0.61	-0.002	-0.002	
	smooth_fraction	0.79	0.97	+0.013	-0.029	1.6	0.24	0.77	-0.005	-0.007	
	stellar_mass	0.98	1.00	+0.022	-0.032	2.9	0.35	0.64	-0.002	-0.002	

Table 7: Counterfactual scaling of the readout’s in-sphere normal component, 512 anchors per run. Bending as the manifold does helps at nearly every anchor, the mirror-image bending hurts at nearly every anchor, a random normal direction does neither; $t^* > 1$ throughout, so the ridge-shrunk normal component undershoots.